

ARAVIND THUMSI JANAKIRAMAN

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Bringing over 3 years of expertise in software development lifecycle, application design, and scalable systems, with a demonstrated proficiency in crafting low-latency programs that prioritize minimal computing resources. Thorough understanding of Web Development, Databases, Operating Systems, with programming knowledge in **Java, Python, C++, C , SQL, HTML and JavaScript** and working in **Agile** environment.

SKILLS SUMMARY

Languages - Java, C++, Python, C, Shell, Linux, Unix , HTML, CSS

Technologies – TIBCO BWCE , MFT, Kafka, Git, AWS, REST APIs, SQL, NoSQL, ELK, Docker, Kubernetes

Skills - Backend, Agile, SCRUM, CI/CD, SDLC, Oracle Cloud, GCP, Redis, ETL, CDC , AWS , Azure

EDUCATION

McDaniel College , *Master of Science in Management Science , 4.0/4*

Ongoing

Southern Illinois University, *Master of Science in Computer Science, 3.87/4*

DEC 2021

Amrita Vishwa Vidya Peetham, *Bachelor of Technology, 9.28/10*

May 2018

EXPERIENCE

IT Consultant | Neonsoft Inc.

May 2024 – Present

- Designed and developed scalable web applications using Angular, Spring Boot, TypeScript, and SQL.
- Built and maintained RESTful APIs, integrating with internal systems via JSON, XML, and Kafka.
- Managed SQL Server databases, optimizing performance via stored procedures, indexing, and query tuning.
- Developed dashboards and reports using SSIS, Tableau, Power BI.
- Conducted unit testing with JUnit, NUnit, and improved logging via Log4Net.

Software Engineer II | Capgemini, Inc.

Jan 2022 – Apr 2024

- Led a 3-person team to seamlessly integrate Slack with PagerDuty and Ocean Application, automating incident alerts during customer downtime and bringing down the intervention rate to 2%.
- Optimized Coca-Cola's AWS Data Lake infrastructure, achieving 99.9% uptime and reducing costs by 25%.
- Automated ETL pipelines using AWS Glue, Python, SQL, improving data processing efficiency by 20%.
- Managed TIBCO applications across business functionalities, enhancing system reliability by 30%.
- Utilized Elasticsearch and employed the ELK stack for logging and monitoring systems, integrated Beats for ops.
- Developed inspector daemon with Python REST APIs for upgrading Oracle DBs from 12c to 19c version.
- Conduct daily health checks on applications to ensure system stability and performance, utilizing monitoring tools such as Splunk.
- Collaborated with cross-functional teams to implement a proactive monitoring system, leading to a 30% reduction in critical system failures during peak production hours.
- Handle MER's to enhance the application business logic to reduce bottlenecks or data validation errors.
- Carryout bi-monthly patching on servers across all environments including Production, Pre-Prod, Test and Dev environment.
- Attended everyday Enterprise Workflow Stand-Up meeting to provide status, thus implementing Agile (Scrum) Methodology.
- Mitigated N+1 query issue via eager loading in SQLite DBs, boosting query performance by 40%.
- Developed on-disk local caching in API servers, cutting response times by 40% and network I/O by 30%.
- Engineered core indexing feature, slashing read query times by 35% and elevating application performance.
- Enhanced Oracle DB scalability through the implementation of soft delete, optimizing index management, resulting in a 20% performance improvement, accommodating increasing data demands.

Graduate Teaching Assistant | Southern Illinois University Edwardsville

Jun 2019- Dec 2021

- Assisted in teaching Advanced Database Management & Computer Architecture, mentoring students in SQL, PostgreSQL, MongoDB, and cloud-based storage solutions.
- Designed hands-on assignments and lab sessions using AWS RDS, Azure SQL Database, and Google BigQuery, enhancing students' cloud database skills.
- Led workshops on DevOps for databases, covering CI/CD, Git, automated backups, and cloud security best practices.
- Mentored students on capstone projects, helping design cost-efficient, scalable cloud architectures using AWS

Lambda and Firebase.

- Assisted faculty in research and curriculum development, integrating real-world case studies on cloud migration and security into coursework.

Assistant software engineer | UGAM solutions, India |

Apr 2018-Dec 2018

- Worked on projects dealing with databases and networking technologies.
- Developed and implemented the project Sales Run Rate. It provides the database of distributor, salesman, retailer, and products. A report to view target and completion is achieved as a result through bar graph which generates with statistic record in the system.
- Created sales performance dashboards using statistical visualization.

PROJECTS

Verizon IT consultant | Neonsoft Inc. | May 2024- Present

- Developed and optimized Java-based applications for enterprise systems, ensuring high performance and scalability.
- Managed database systems (Oracle, MySQL, PostgreSQL), optimizing query performance and data integrity.
- Designed and implemented RESTful APIs for seamless data exchange between internal systems.
- Automated database maintenance and ETL workflows, reducing manual efforts and enhancing efficiency.

Entergy Middleware Production Support | Capgemini Inc. | Jan 2023 – Apr 2024

- Improved TIBCO middleware reliability by 30% and reduced resolution time by 25% through automation.
- Managed 50+ TIBCO deployments, ensuring 99.9% uptime across environments.
- Designed and deployed Spring Boot microservices on AWS and Azure for enhanced scalability.
- Automated deployments with Jenkins and PowerShell, reducing manual intervention.
- Optimized cloud databases (AWS RDS, DynamoDB, Redshift, BigQuery) for performance and cost efficiency.

Coca-Cola Data Lake Production Support | Capgemini Inc. | Feb 2022 – Dec 2022

- Maintained 99.9% uptime for Coca-Cola's AWS Data Lake, resolving 20+ critical issues per month.
- Developed real-time monitoring solutions using AWS CloudWatch, cutting system downtime by 25%.
- Improved data analytics accessibility via AWS Athena queries and QuickSight dashboards.
- Strengthened security compliance by configuring CloudTrail and AWS Config for automated monitoring

ACADEMIC PROJECTS

Memory Optimization: Transparent Superpage Support - Report

- Reduced TLB misses by 10% and improved efficiency by aligning super pages with virtual address space.
- Performed simulations using memory traces from Apache Web Server Benchmarks, demonstrating a decrease in TLB misses with frequent promotions and demotions of superpages.

Covid-19 Resource Prediction - Report

- Established an 80% correlation between rising COVID cases and severity of online substance abuse posts.
- Applied ARIMA model for time series analysis of hospital bed availability, yielding 90%+ prediction accuracy.

DNS Resolver with DNSSEC Integration

- Built a custom DNS resolver capable of iteratively querying root servers to name servers to resolve DNS queries.
- Implemented DNS resolver with DNSSEC, demonstrating public-private key cryptography principles.

Raft Consensus Implementation for Distributed Key-Value Store

- Ensured data replication and leader election mechanisms to maintain system availability in the presence of node failures. Implemented log-based approach for tracking and synchronizing state changes across distributed nodes.

Broadridge Innovation Lab: AI-Driven ESG Analytics and Solutions Development

- Conducted data extraction and analysis of ESG ratings and disclosures for 4,400+ US and Canadian companies.

Identified industry trends, disclosure patterns, and performance benchmarks among top and bottom quartile companies.

CERTIFICATIONS

- **AWS Solution Architect Associate**
- **AWS Certified Cloud Practitioner**
- **Agile Software Development – University of Minnesota.**
- **IBM Data science**
- **Oracle Certified Professional, Java SE 8 Programmer**

WORKSHOP / HACKATHONS

- **Smart India Hackathon 2018 3rd place representing Amrita Vishwa Vidyapeetham.**
- **ANOKHA 2017, ANOKHA 2018 - Techfest at Amrita, Tech event's organizer.**